

Detecting Student Engagement in Classrooms for Intelligent Tutoring Systems

Hoang Tieu Binh
Hanoi National University of Education
Hanoi, Vietnam
Email: htbinh@hnue.edu.vn

Nguyen Quang Trung
Vietnam Youth Academy
Hanoi, Vietnam
Email: trungnq@vya.edu.vn

Hoang-Anh The Nguyen
Institute of Information Technology
Vietnam Academy of Science and Technology
Email: nt.hoanganh@ioit.ac.vn

Bui The Duy
University of Engineering and Technology
Hanoi, Vietnam
Email: duybt@vnu.edu.vn

ABSTRACT

Detecting and tracking student engagement in a large classroom can help tutors or automatic learning systems easily control or summarize the situation. To come up with the advanced technique in machine learning, especially deep learning, nowadays, many schools can build an efficient system for supporting teachers or tutoring systems. In this paper, we propose a transfer learning method applying to a small dataset to classify student actions in the classroom. Another contribution is building a lightweight dataset with a limited number of images for each category for classification work. The experiments show the acceptable result in action recognition with high accuracy compared to other researches.

Keywords: Convolutional neural networks, transfer learning, student engagement, student actions classification, intelligent tutoring systems.

I. INTRODUCTION

In recent years, detecting student emotion and actions in the classroom is a key topic for researchers who focus on intelligent tutoring systems (ITSs). Together with the development of advanced techniques and the power of computing ability, many solutions for these problems are presented and gained some positive results. Knowing exactly whether a student is engaged in the classroom is very important for designing an adaptive learning system and intelligent tutoring systems. Besides this, detecting student engagement supports teachers (in a traditional classroom) or tutoring systems in tracking learner attention. The success of a traditional course or online learning course depends on the outcomes of students, and that results are related to the engagement of the learners. Some researches

show that engagement detection makes opportunities for improving learning and teaching [1], [2] (such as adjusting teaching strategy the way good teachers do, getting real-time feedback about the level of engagement of audiences) and even for issuing targeted interventions of the monitors or educational manager [3].

In the firm of engagement detection, according to Dewan [4], he divided engagement detection into their categories, such as Automatic, Semi-Automatic and Manual. Toward computer vision-based point of view, he mentioned that Facial Expression, Gesture, and posture, Eye movement are three facets of automatic detection.

Facial expression recognition receives many concerns from researchers. Many researches concentrate on student's face to guess the emotions of learners. Nezami *et al.* [5] used a convolutional neural network to classify the emotion of students, based on the pre-trained result from FER-2013 dataset [6]. Whitehill *et al.* [3] divided student engagement into four levels from 1 to 4 and used the support vector machine (SVM) classifier to detect engagement. Bosch in [7] focused on mind wandering - a type of cognitive disengagement. He concentrated on extracting low-level and high-level features from video clips for the face recognition process.

Eye movement is interested in recent years, Raina *et al.* [8] proposed an eye-tracking-based model which used eye-tracker equipment to compare engagement between two student groups to reduce content skipping when they study on the online environment. Krithika and Lakshmi [9] built a system call SERS (Student Emotion Recognition System) that can identify and monitor the emotions of the student in an e-learning environment and provide a real-time feedback mechanism to enhance the e-learning aids for better content delivery. The system can detect eyes, head movement for

estimating student engagement level.

Our work focuses on the third branch: **Gesture and postures** because this kind of non-verbal communication can express various student states in the classroom. Gesture and postures are two important forms of non-verbal communication through our body language [4].

In this paper, we deal with engagement during learning via technology. Some hidden cameras are set up to collect images in classrooms in learning time. Then we build a dataset for classifying student actions in the classroom to identify whether a student is interested in learning or not. We use a convolutional neural network to train my new data. Before this, the network was pre-trained to get the standard weighted matrix for ordinary actions.

The paper is organized as follows. Firstly, the dataset that is used in this paper is introduced in Sec. III. This is followed by the motivation for the classification of student actions in Sec. IV. Results and discussion are subsequently presented in Sec. V, after which the paper is concluded in Sec. VI

II. RELATED WORK

Many computer-based approaches have been used to detect engagements with gesture and posture information. In 2011, Fang et al. [10] recognized six popular actions in the classroom such as "raising the left hand", "raising the right hand", "raising two hands", "standing up", "lying prone", "and normal posture" to help the teacher notice some behaviors of the students in teaching time. Grafsgaard et al. [11] measured the postural quantity of motion of hands and head to find the relationship between nonverbal behaviors and dialogue and its effects while learning. Potnis and Jahagirdar [12] captured videos and detected hand gestures in the classroom to convey it to the lecturer. Sathyanarayana et al. [13] built a dataset called SDMATH and used HOG (histograms of oriented gradients) and SVM (support vector machines) to localize deictic gestures in a teaching-learning session. Zaletelj and Košir [14] measured students' attention level in the classroom by using Kinect device. They combined upper body posture, face gaze points, and facial features and used seven classifiers for classification, then compared the results between these methods. Rich *et al.* [15] used gesture and speech as one channel to follow up the user engagement during human-robot interactions. Klein and Celik [16] built a system called WITS (Wits Intelligent Teaching System) to support lecturers to track students' engagement by monitoring their real-time feedback. They created an Interesting Map which pointed out students who are engaged or disengaged in the classroom in the lecture time.

Table I shows some related up-to-date researches which were carried out by using posture and gesture features such as hand gesture, body posture, finger movement. Some

popular data mining methods were also conducted, such as convolutional neural network, SVM, KNNs...Video formats were preferable to image in these works.

III. DATA COLLECTION

Related to this paper, we conducted a data collection from several classes in the lecture time. We built a database by extracting images from high-density videos (HD quality) which were captured by some fixed angle cameras in the classroom. These cameras were secret to the learners to ensure the natural behavior of the students. According to popular postures of a student in the lecture time, several actions can be considered as main actions and they were labeled as *interest (engagement)* and *non-interest (disengagement)*. Data after collecting was handily pre-processed as cropping, reducing resolution and manually labeling.

The dataset includes 942 files which are divided into 8 classes in 2 groups. The first group is labeled as interest (engagement) and non-interest (disengagement) for the others. With the interest group, we continuously divided into some actions such as "writing", "reading", "raising hand", "looking forward". On the other hand, with the non-interest group, we split into "using telephone", "looking up", "looking around", "head down".

The labeled data was divided into two sets, training set and test set with the rate of 80% and 20%, respectively.



Fig. 1: List of student actions in classroom (left to right, top to bottom): "writing", "reading", "raising hand", "looking forward" and "using telephone", "looking up", "looking around", "head down". 4 images on the top row represent for interesting state, and the others for non-interesting.

IV. EXPERIMENT

In recent years, deep learning shows some advantages among other learning methods and artificial intelligence. Especially, Convolutional neural network is one of the deep structure with archive state-of-the-art result in various domains. The main components of this network include

Name	Data type	Features	Method
Fang <i>et al.</i> [10]	Video sequences	Gesture	Object segmentation
Klein and Celik [16]	Image sequences	Posture and gesture	Convolutional Neural Network
Grafsgaard <i>et al.</i> [11]	Kinect depth video	Posture and gesture	Regression
Potnis and Jahagirdar [12]	Images	Hand gesture	Bag-of-Features, SVM
Sathyanarayana <i>et al.</i> [13]	Video, audio	Speech, gestures, eye gaze, facial expression	HOG, SVM
Zaletelj and Košir [14]	Kinect video	Body posture, face gaze, facial features	Bagged trees, KNNs

TABLE I: Previous works for engagement detection in classroom based on posture and gesture features

some convolutional layers (sub-sampling layers), Max-pooling layers, Fully-connected layers and a soft-max layer to identify the number of classes. In this paper, we construct

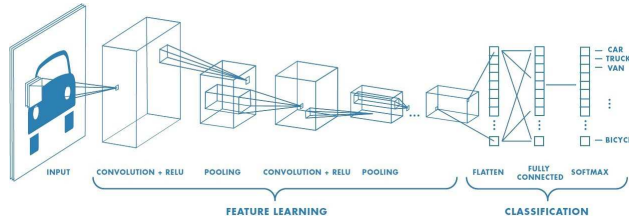


Fig. 2: A CNN structure sample from Toward Data Science [17]. Accessed on June, 2019

a convolutional neural network with the input is a series of 224x224 images which are resized from the origin. First, we re-use deep convolutional network based on 2 structures VGG16 and VGG19 [18]. The current network is also pre-trained on the ImageNet [19] dataset which contains 1.2 million high-resolution images [20] for speeding up convergence. Because of the small dataset, we apply transfer learning [21] and use data augmentation techniques as in [22] to improve overall accuracy and avoid overfitting. Some augmentation techniques like rescaling, flipping, cropping, rotating and shifting are applied. Learning rate is set to 0.0001 and momentum of the network is 0.9 (based on default values of Keras library). Secondly, the first trained model is pop out by freezing some first layers (first 5 layers are excluded from training) for the next classification process. Next steps, we use regularization technique with dropout rate [23] is 0.2. Finally, 8 classes which are equivalent to 8 actions in the classroom are classified by a softmax activation function. Our proposed procedure was mentioned as below:

- Step 1: Create a new model which is inherited from the architecture of VGG.
- Step 2: Freeze some first layers of the new model.
- Step 3: Use weights matrix from ImageNet.
- Step 4: Use data augmentation with new dataset.
- Step 5: Re-compile and continue to train pre-trained model with small target dataset.

Our experiment is conducted on an Intel Xeon based system with an 8Gb memory GPU card GeForce GTX 1070 and

CUDA 9.0 library. Keras framework [24] with TensorFlow backend is chosen for developing our convolutional network. The structure of this network is represented as Fig. 3.

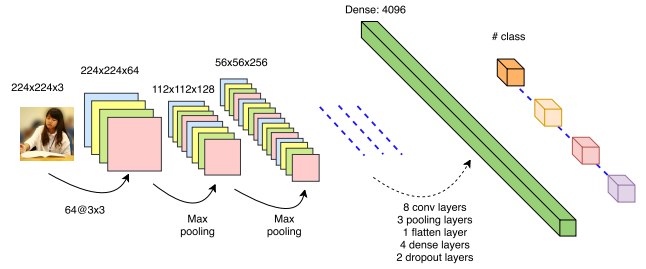


Fig. 3: Convolutional neural network architecture. First layers are based on VGG16 and VGG19 structures. Five top layers are retrained with new parameters.

V. RESULTS

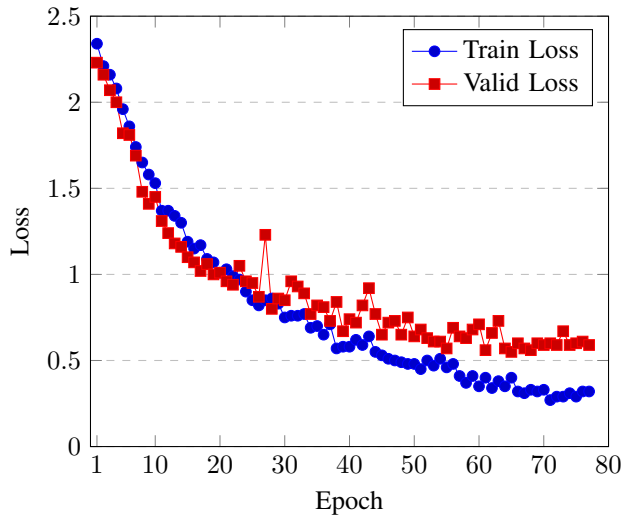
Although the comparison between researches is not appropriate because of the difference of the datasets and methods, we just show the accuracy of the other similar papers to prove that our achievement is acceptable when applying very lightweight dataset. In our experiment, our accuracy gained 80.8% compared to [10] gained 72% . Klein [16] showed impressive result with the accuracy was 89.7% after 20,000 training epochs and over 200,000 images.

Fig 4a and 4b represent for Loss curves and Accuracy

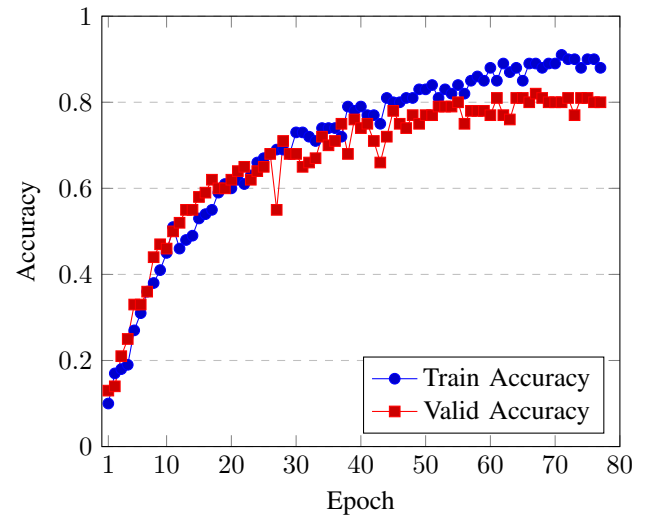
Method	Accuracy rate
Fang [10]	72.0%
Zaletelj <i>et al.</i> [14]	75.3%
Training CNN from scratch	68.8%
Training CNN with VGG16	74.9%
Training CNN with VGG19	76.0%
Training CNN with VGG19 and applied transfer learning	80.8%

TABLE II: Accuracy between some methods. Our method shows some encouraging results in action recognition in classroom.

curves, respectively. After approximate 80 iterations, the



(a) Loss curves of training and validating process.



(b) Accuracy curves of training and validating process.

Fig. 4: The process was early stopped since no improvement in test accuracy for a long time of period. We used early stopping to prevent overfitting. The training curves and validation curves are similar which illustrated that the model is not overfitting.

validating loss/accuracy does not decrease/increase. The early stopping trigger activate the immediate stopping process to prevent model overfitting.

One of the reasons why the result is not high enough compared to [16] because of the ambiguity of actions of the student. The actions "head down" and "reading" are too similar cause the CNNs recognize as the same action (Fig. 5).

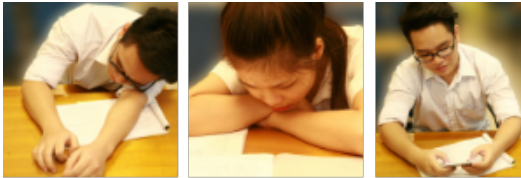


Fig. 5: Ambiguous images from the dataset which wrongly recognized. From left to right: "Head down" was recognized as "Reading", "Reading" was recognized as "Head down", "Using telephone" was recognized as "Reading".

Actual	Predicted	
	Engagement	Disengagement
Engagement	51	9
Disengagement	14	46

TABLE III: Convolutional Neural Network confusion matrix for the test set.

VI. CONCLUSION AND FUTURE WORK

Posture and gesture are fundamental components of embodied affect which are tightly related to students' state in

the classroom. Toward the next generation of intelligent tutoring systems, the ability of analysis correctly these states may help to support both students and tutors.

In this paper, we have concretely described an approach to detect students' engagement based on their gesture and posture in the classroom. These illustrate that applying transfer learning and convolutional neural network can classify effectively student actions in the classroom based on pre-defined actions labels. In the future, to improve the overall accuracy of the classification process, we make a plan to expand the dataset by collecting more images. Besides this, we can build more convolutional neural networks to form multi-streams CNN for incorporating complementary features of motion networks [25]. We can also use vote fusion method which mentioned by Binh *et al.* [26] to optimize recognition process.

Last but not least, the results of this recognition process is only the first step of an intelligent tutoring system. To support teachers in the classroom or some of the e-learning environments, we need to build more components such as domain knowledge model, student trait model, human-machine interface and other necessary modules for a complete intelligent tutoring system.

ACKNOWLEDGMENTS

This work is supported by National Key Laboratory of Networking and Multimedia Technology (Grant No. PT-NTD19.03) and FIRST Central Project Management Unit, Ministry of Science and Technology, Vietnam under Grant Agreement number 12/FIRST/2a/IoIT.

REFERENCES

- [1] J. Smallwood, D. J. Fishman, and J. W. Schooler, "Counting the cost of an absent mind: Mind wandering as an underrecognized influence on educational performance," *Psychonomic Bulletin & Review*, vol. 14, no. 2, pp. 230–236, Apr. 2007.
- [2] S. D'Mello, "A selective meta-analysis on the relative incidence of discrete affective states during learning with technology," *Journal of Educational Psychology*, vol. 105, no. 4, pp. 1082–1099, 2013.
- [3] J. Whitehill, Z. Serpell, Y.-C. Lin, A. Foster, and J. R. Movellan, "TRANSACTIONS ON AFFECTIVE COMPUTING The Faces of Engagement: Automatic Recognition of Student Engagement from Facial Expressions," vol. 5, no. 1, pp. 86–98, 2014.
- [4] M. A. A. Dewan, M. Murshed, and F. Lin, "Engagement detection in online learning: a review," *Smart Learning Environments*, vol. 6, no. 1, Jan. 2019.
- [5] O. M. Nezami, L. Hamey, D. Richards, and M. Dras, "Deep Learning for Domain Adaption: Engagement Recognition," *CoRR*, vol. abs/1808.02324, 2018.
- [6] I. J. Goodfellow, D. Erhan, P. L. Carrier, A. Courville, M. Mirza, B. Hamner, W. Cukierski, Y. Tang, D. Thaler, D.-H. Lee, Y. Zhou, C. Ramaiah, F. Feng, R. Li, X. Wang, D. Athanasakis, J. Shawe-Taylor, M. Milakov, J. Park, R. Ionescu, M. Popescu, C. Grozea, J. Bergstra, J. Xie, L. Romaszko, B. Xu, Z. Chuang, and Y. Bengio, "Challenges in representation learning: A report on three machine learning contests," *Neural Networks*, vol. 64, pp. 59–63, apr 2015.
- [7] N. Bosch, "Detecting Student Engagement : Human Versus Machine," *24th Conf. User Model. Adapt. Pers. (UMAP 2016)*, pp. 317–320, 2016.
- [8] S. Raina, L. Bernard, B. Taylor, and S. Kaza, "Using eye-tracking to investigate content skipping: A study on learning modules in cybersecurity," in *2016 IEEE Conference on Intelligence and Security Informatics (ISI)*. IEEE, Sep. 2016. [Online]. Available: <https://doi.org/10.1109/isi.2016.7745486>
- [9] K. L.B and L. Priya GG, "Student Emotion Recognition System (SERS) for e-learning Improvement Based on Learner Concentration Metric," *Procedia Computer Science*, vol. 85, pp. 767–776, 12 2016.
- [10] C.-Y. Fang, M.-H. Kuo, G.-C. Lee, and S.-W. Chen, "Student gesture recognition system in classroom 2.0," in *Computers and Advanced Technology in Education*. ACTAPRESS, 2011.
- [11] J. F. Graßgaard, J. B. Wiggins, K. E. Boyer, E. N. Wiebe, and J. C. Lester, "Embodied affect in tutorial dialogue: Student gesture and posture," in *Lecture Notes in Computer Science*. Springer Berlin Heidelberg, 2013, pp. 1–10.
- [12] S. S. Potnis, , and A. S. Jahagirdar, "Real time hand gesture recognition for smart classroom environment," *International Journal of Computer Trends and Technology*, vol. 17, no. 2, pp. 78–83, Nov. 2014.
- [13] S. Sathayanarayana, R. K. Satzoda, A. Carini, M. Lee, L. Salamanca, J. Reilly, D. Forster, M. Bartlett, and G. Littlewort, "Towards automated understanding of student-tutor interactions using visual deictic gestures," in *2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops*. IEEE, Jun. 2014.
- [14] J. Zaletelj and A. Košir, "Predicting students' attention in the classroom from Kinect facial and body features," *EURASIP Journal on Image and Video Processing*, vol. 2017, no. 1, Dec. 2017.
- [15] C. Rich, B. Ponsleur, A. Holroyd, and C. L. Sidner, "Recognizing engagement in human-robot interaction," *Proceeding of the 5th ACM/IEEE international conference on Human-robot interaction - HRI '10*, no. March, p. 375, 2010.
- [16] R. Klein and T. Celik, "The WITS intelligent teaching system: Detecting student engagement during lectures using convolutional neural networks," in *2017 IEEE International Conference on Image Processing (ICIP)*. IEEE, Sep. 2017.
- [17] CNN, "Towards Data Science," <https://towardsdatascience.com/>, accessed: 2019-June-10.
- [18] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," 2014. [Online]. Available: <http://arxiv.org/abs/1409.1556>
- [19] J. Deng, R. Socher, L. Fei-Fei, W. Dong, K. Li, and L.-J. Li, "ImageNet: A large-scale hierarchical image database," in *2009 IEEE Conference on Computer Vision and Pattern Recognition(CVPR)*, 06 2009, pp. 248–255.
- [20] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *Advances In Neural Information Processing Systems*, pp. 1–9, 2012.
- [21] L. Torrey and J. Shavlik, "Transfer Learning," *Encyclopedia of the Sciences of Learning*, pp. 3337–3337, 2012.
- [22] L. Wang, Y. Xiong, Z. Wang, Y. Qiao, D. Lin, X. Tang, and L. V. Gool, "Temporal segment networks: Towards good practices for deep action recognition," in *Computer Vision – ECCV 2016*. Springer International Publishing, 2016, pp. 20–36.
- [23] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014. [Online]. Available: <http://jmlr.org/papers/v15/srivastava14a.html>
- [24] F. Chollet *et al.*, "Keras," <https://keras.io>, 2015.
- [25] Z. Tu, W. Xie, Q. Qin, R. Poppe, R. C. Veltkamp, B. Li, and J. Yuan, "Multi-stream CNN: Learning representations based on human-related regions for action recognition," *Pattern Recognition*, vol. 79, pp. 32–43, Jul. 2018. [Online]. Available: <https://doi.org/10.1016/j.patcog.2018.01.020>
- [26] H. T. Binh, M. T. Chau, A. Sugimoto, and B. T. Duy, "Selecting active frames for action recognition with vote fusion method," in *2018 7th International Conference on Computer and Communication Engineering (ICCCE)*. IEEE, sep 2018.